8 Solution Evaluation

The Solution Evaluation knowledge area describes the tasks that business analysts perform to assess the performance of and value delivered by a solution in use by the enterprise, and to recommend removal of barriers or constraints that prevent the full realization of the value.

While there may be some similarities to the activities performed in Strategy Analysis (p. 99), or Requirements Analysis and Design Definition (p. 133), an important distinction between the Solution Evaluation knowledge area and other knowledge areas is the existence of an actual solution. It may only be a partial solution, but the solution or solution component has already been implemented and is operating in some form. Solution Evaluation tasks that support the realization of benefits may occur before a change is initiated, while current value is assessed, or after a solution has been implemented.

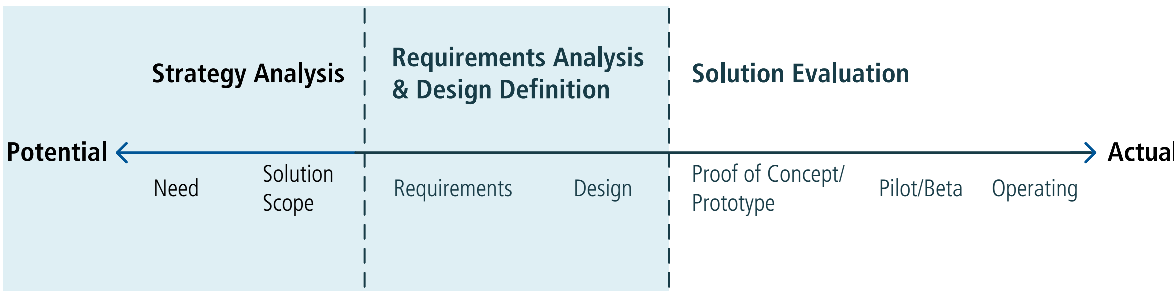
Solution Evaluation tasks can be performed on solution components in varying stages of development:

- **Prototypes or Proofs of Concept**: working but limited versions of a solution that demonstrate value.
- **Pilot or Beta releases**: limited implementations or versions of a solution used in order to work through problems and understand how well it actually delivers value before fully releasing the solution.
- **Operational releases**: full versions of a partial or completed solution used to achieve business objectives, execute a process, or fulfill a desired outcome.

Solution Evaluation describes tasks that analyze the actual value being delivered, identifies limitations which may be preventing value from being realized, and makes recommendations to increase the value of the solution. It may include any combination of performance assessments, tests, and experiments, and may combine both objective and subjective assessments of value. Solution Evaluation generally focuses on a component of an enterprise rather than the entire enterprise.

The following image illustrates the spectrum of value as business analysis activities progress from delivering potential value to actual value.

Figure 8.0.1: Business Analysis Value Spectrum



The Solution Evaluation knowledge area includes the following tasks:

- **Measure Solution Performance**: determines the most appropriate way to assess the performance of a solution, including how it aligns with enterprise goals and objectives, and performs the assessment.
- **Analyze Performance Measures**: examines information regarding the performance of a solution in order to understand the value it delivers to the enterprise and to stakeholders, and determines whether it is meeting current business needs.
- **Assess Solution Limitations**: investigates issues within the scope of a solution that may prevent it from meeting current business needs.
- **Assess Enterprise Limitations**: investigates issues outside the scope of a solution that may be preventing the enterprise from realizing the full value that a solution is capable of providing.
- **Recommend Actions to Increase Solution Value**: identifies and defines actions the enterprise can take to increase the value that can be delivered by a solution.

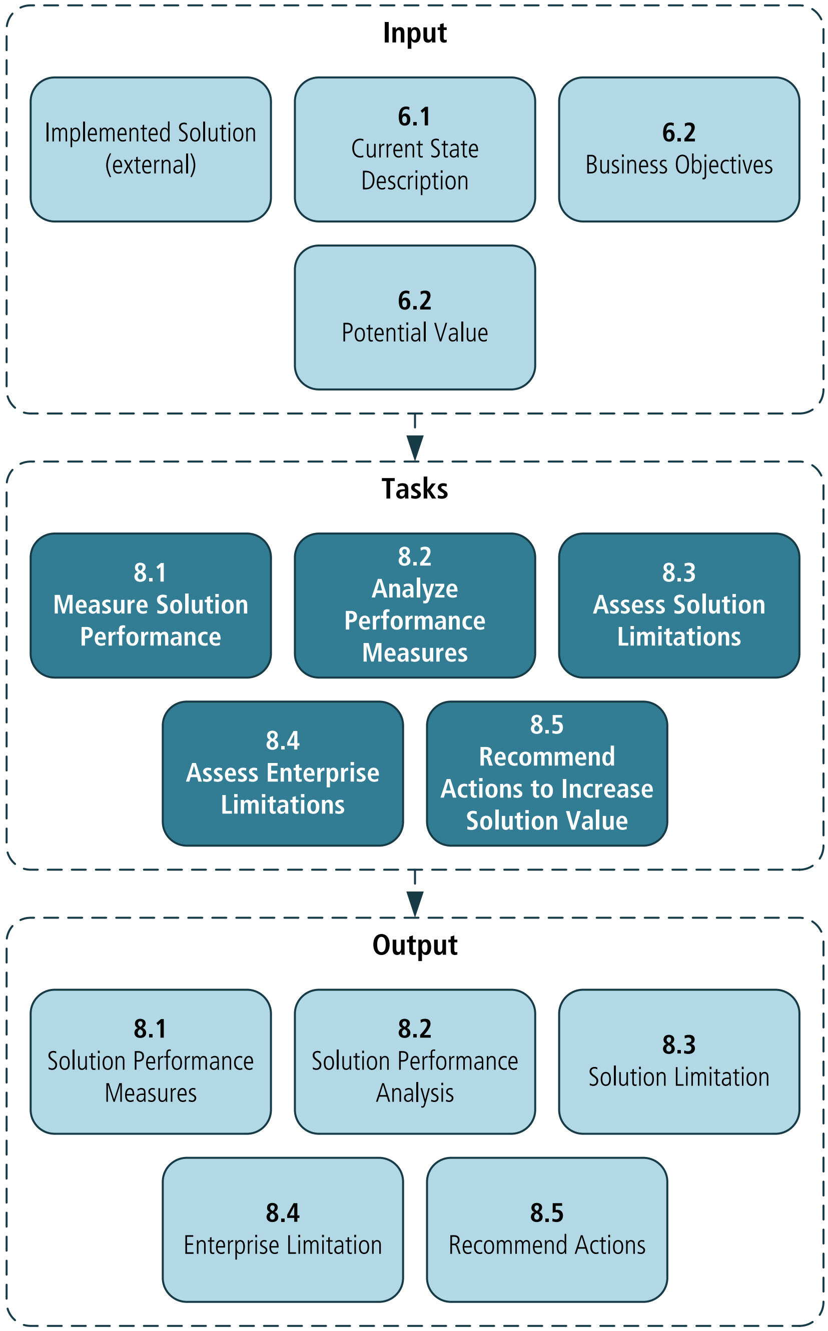
The Core Concept Model in Solution Evaluation

The *Business Analysis Core Concept Model*™ (*BACCM*™) describes the relationships among the six core concepts. The following table describes the usage and application of each of the core concepts within the context of Solution Evaluation.

Table 8.0.1: : The Core Concept Model in Solution Evaluation

Core Concept	During Solution Evaluation, business analysts...
Change: the act of transformation in response to a need.	recommend a change to either a solution or the enterprise in order to realize the potential value of a solution.
Need: a problem or opportunity to be addressed.	evaluate how a solution or solution component is fulfilling the need.
Solution: a specific way of satisfying one or more needs in a context.	assess the performance of the solution, examine if it is delivering the potential value, and analyze why value may not be realized by the solution or solution component.
Stakeholder: a group or individual with a relationship to the change, the need, or the solution.	elicit information from the stakeholders about solution performance and value delivery.
Value: the worth, importance, or usefulness of something to a stakeholder within a context.	determine if the solution is delivering the potential value and examine why value may not be being realized.
Context: the circumstances that influence, are influenced by, and provide understanding of the change.	consider the context in determining solution performance measures and any limitations within the context that may prohibit value from being realized.

Figure 8.0.2: Solution Evaluation Input/Output Diagram



8.1 Measure Solution Performance

8.1.1 Purpose

The purpose of Measure Solution Performance is to define performance measures and use the data collected to evaluate the effectiveness of a solution in relation to the value it brings.

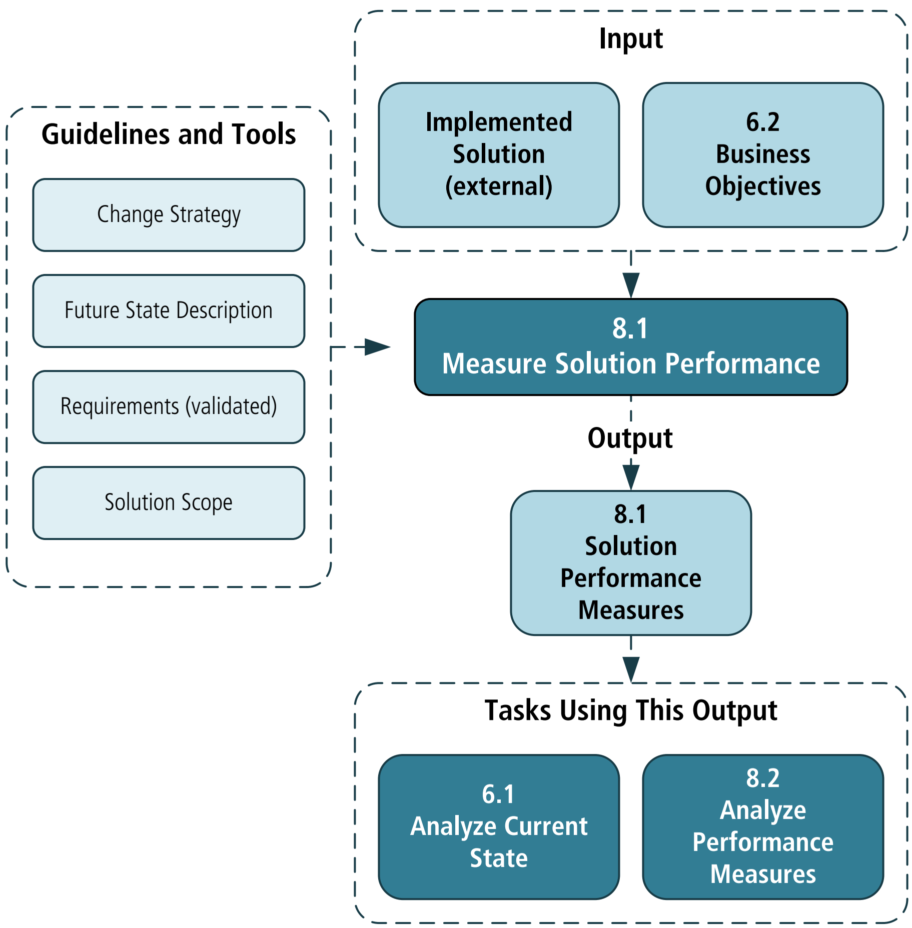
8.1.2 Description

Performance measures determine the value of a newly deployed or existing solution. The measures used depend on the solution itself, the context, and how the organization defines value. When solutions do not have built-in performance measures, the business analyst works with stakeholders to determine and collect the measures that will best reflect the performance of a solution. Performance may be assessed through key performance indicators (KPIs) aligned with enterprise measures, goals and objectives for a project, process performance targets, or tests for a software application.

8.1.3 Inputs

- **Business Objectives**: the measurable results that the enterprise wants to achieve. Provides a benchmark against which solution performance can be assessed.
- **Implemented Solution (external)**: a solution (or component of a solution) that exists in some form. It may be an operating solution, a prototype, or a pilot or beta solution.

Figure 8.1.1: Measure Solution Performance Input/Output Diagram



8.1.4 Elements

.1 Define Solution Performance Measures

When measuring solution performance, business analysts determine if current measures exist, or if methods for capturing them are in place. Business analysts ensure that any existing performance measures are accurate, relevant and elicit any additional performance measures identified by stakeholders.

Business goals, objectives, and business processes are common sources of measures. Performance measures may be influenced or imposed by third parties such as solution vendors, government bodies, or other regulatory organizations. The type and nature of the measurements are considered when choosing the elicitation method. Solution performance measures may be quantitative, qualitative, or both, depending on the value being measured.

- **Quantitative Measures**: are numerical, countable, or finite, usually involving amounts, quantities, or rates.
- **Qualitative Measures**: are subjective and can include attitudes, perceptions, and any other subjective response. Customers, users, and others involved in the operation of a solution have perceptions of how well the solution is meeting the need.

.2 Validate Performance Measures

Validating performance measures helps to ensure that the assessment of solution performance is useful. Business analysts validate the performance measures and any influencing criteria with stakeholders. Specific performance measures should align with any higher-level measures that exist within the context affecting the solution. Decisions about which measures are used to evaluate solution performance often reside with the sponsor, but may be made by any stakeholder with decision-making authority.

.3 Collect Performance Measures

When defining performance measures, business analysts may employ basic statistical sampling concepts.

When collecting performance measures, business analysts consider:

- **Volume or Sample Size**: a volume or sample size appropriate for the initiative is selected. A sample size that is too small might skew the results and lead to inaccurate conclusions. Larger sample sizes may be more desirable, but may not be practical to obtain.
- **Frequency and Timing**: the frequency and timing with which measurements are taken may have an effect on the outcome.
- **Currency**: measurements taken more recently tend to be more representative than older data.

Using qualitative measures, business analysts can facilitate discussions to estimate the value realized by a solution. Stakeholders knowledgeable about the operation and use of the solution reach a consensus based on facts and reasonable assumptions, as perceived by them.

8.1.5 Guidelines and Tools

- **Change Strategy**: the change strategy used or in use to implement the potential value.
- **Future State Description**: boundaries of the proposed new, removed, or modified components of the enterprise, and the potential value expected from the future state.
- **Requirements (validated)**: a set of requirements that have been analyzed and appraised to determine their value.
- **Solution Scope**: the solution boundaries to measure and evaluate.

8.1.6 Techniques

- **Acceptance and Evaluation Criteria**: used to define acceptable solution performance.
- **Benchmarking and Market Analysis**: used to define measures and their acceptable levels.
- **Business Cases**: used to define business objectives and performance measures for a proposed solution.
- **Data Mining**: used to collect and analyze large amounts of data regarding solution performance.
- **Decision Analysis**: used to assist stakeholders in deciding on suitable ways to measure solution performance and acceptable levels of performance.
- **Focus Groups**: used to provide subjective assessments, insights, and impressions of a solution's performance.
- **Metrics and Key Performance Indicators (KPIs)**: used to measure solution performance.
- **Non-Functional Requirements Analysis**: used to define expected characteristics of a solution.
- **Observation**: used either to provide feedback on perceptions of solution performance or to reconcile contradictory results.
- **Prototyping**: used to simulate a new solution so that performance measures can be determined and collected.
- **Survey or Questionnaire**: used to gather opinions and attitudes about solution performance. Surveys and questionnaires can be effective when large or disparate groups need to be polled.
- **Use Cases and Scenarios**: used to define the expected outcomes of a solution.
- **Vendor Assessment**: used to assess which of the vendor's performance measures should be included in the solution's performance assessment.

8.1.7 Stakeholders

- **Customer**: may be consulted to provide feedback on solution performance.
- **Domain Subject Matter Expert**: a person familiar with the domain who can be consulted to provide potential measurements.
- **End User**: contributes to the actual value realized by the solution in terms of solution performance. They may be consulted to provide reviews and feedback on areas such as workload and job satisfaction.
- **Project Manager**: responsible for managing the schedule and tasks to perform the solution measurement. For solutions already in operation, this role may not be required.
- **Sponsor**: responsible for approving the measures used to determine solution performance. May also provide performance expectations.
- **Regulator**: an external or internal group that may dictate or prescribe constraints and guidelines that must be incorporated into solution performance measures.

8.1.8 Outputs

- **Solution Performance Measures**: measures that provide information on how well the solution is performing or potentially could perform.

8.2 Analyze Performance Measures

8.2.1 Purpose

The purpose of Analyze Performance Measures is to provide insights into the performance of a solution in relation to the value it brings.

8.2.2 Description

The measures collected in the task Measure Solution Performance (p. 166) often require interpretation and synthesis to derive meaning and to be actionable. Performance measures themselves rarely trigger a decision about the value of a solution.

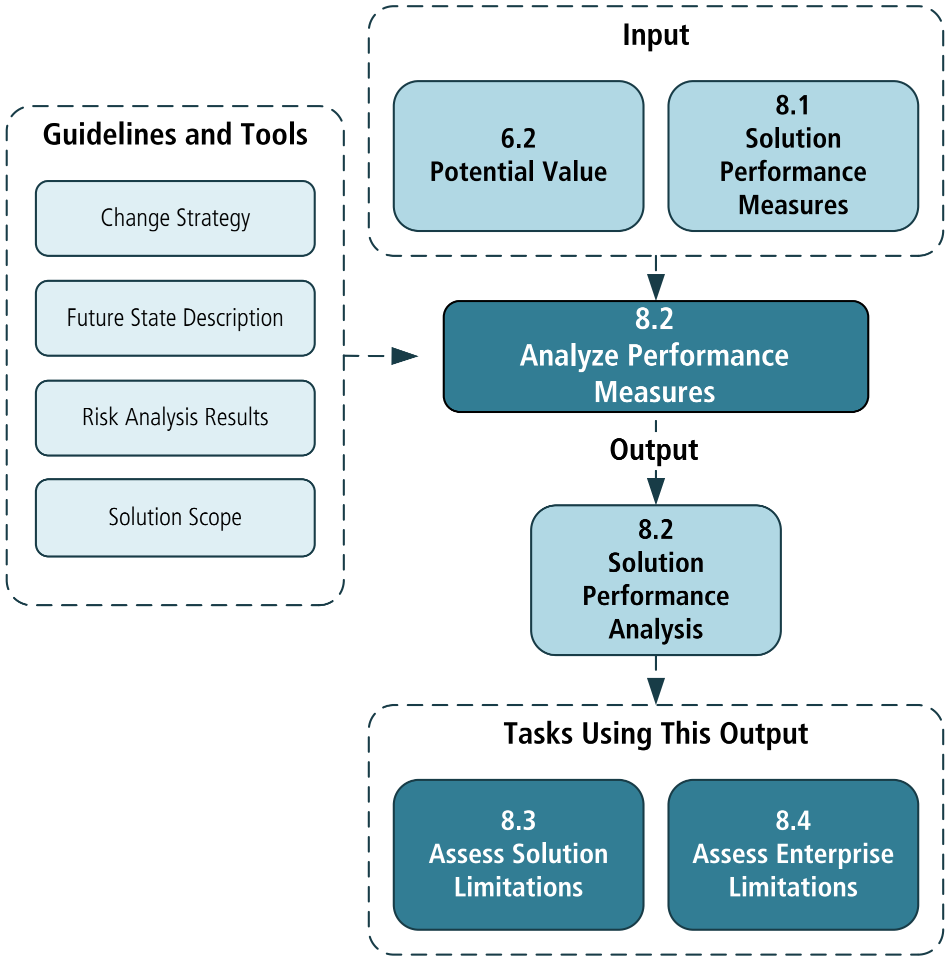
In order to meaningfully analyze performance measures, business analysts require a thorough understanding of the potential value that stakeholders hope to achieve with the solution. To assist in the analysis, variables such as the goals and objectives of the enterprise, key performance indicators (KPIs), the level of risk of the solution, the risk tolerance of both stakeholders and the enterprise, and other stated targets are considered.

8.2.3 Inputs

- **Potential Value**: describes the value that may be realized by implementing the proposed future state. It can be used as a benchmark against which solution performance can be evaluated.

- **Solution Performance Measures**: measures and provides information on how well the solution is performing or potentially could perform.

Figure 8.2.1: Analyze Performance Measures Input/Output Diagram



8.2.4 Elements

.1 Solution Performance versus Desired Value

Business analysts examine the measures previously collected in order to assess their ability to help stakeholders understand the solution's value. A solution might be high performing, such as an efficient online transaction processing system, but contributes lower value than expected (or compared to what it had contributed in the past). On the other hand, a low performing but potentially valuable solution, such as a core process that is inefficient, can be enhanced to increase its performance level. If the measures are not sufficient to help stakeholders determine solution value, business analysts either collect more measurements or treat the lack of measures as a solution risk.

.2 Risks

Performance measures may uncover new risks to solution performance and to the enterprise. These risks are identified and managed like any other risks.

.3 Trends

When analyzing performance data, business analysts consider the time period when the data was collected to guard against anomalies and skewed trends. A large enough sample size over a sufficient time period will provide an accurate depiction of solution performance on which to make decisions and guard against false signals brought about by incomplete data. Any pronounced and repeated trends, such as a noticeable increase in errors at certain times or a change in process speed when volume is increased, are noted.

.4 Accuracy

The accuracy of performance measures is essential to the validity of their analysis. Business analysts test and analyze the data collected by the performance measures to ensure their accuracy. To be considered accurate and reliable, the results of performance measures should be reproducible and repeatable.

.5 Performance Variances

The difference between expected and actual performance represents a variance that is considered when analyzing solution performance. Root cause analysis may be necessary to determine the underlying causes of significant variances within a solution. Recommendations of how to improve performance and reduce any variances are made in the task Recommend Actions to Increase Solution Value (p. 182).

8.2.5 Guidelines and Tools

- **Change Strategy**: the change strategy that was used or is in use to implement the potential value.
- **Future State Description**: boundaries of the proposed new, modified, or removed components of the enterprise and the potential value expected from the future state.
- **Risk Analysis Results**: the overall level of risk and the planned approach to modifying the individual risks.
- **Solution Scope**: the solution boundaries to measure and evaluate.

8.2.6 Techniques

- **Acceptance and Evaluation Criteria**: used to define acceptable solution performance through acceptance criteria. The degree of variance from these criteria will guide the analysis of that performance.
- **Benchmarking and Market Analysis**: used to observe the results of other organizations employing similar solutions when assessing risks, trends, and variances.
- **Data Mining**: used to collect data regarding performance, trends, common issues, and variances from expected performance levels and understand patterns and meaning in that data.

- **Interviews**: used to determine expected value of a solution and its perceived performance from an individual or small group's perspective.
- **Metrics and Key Performance Indicators (KPIs)**: used to analyze solution performance, especially when judging how well a solution contributes to achieving goals.
- **Observation**: used to observe a solution in action if the data collected does not provide definitive conclusions.
- **Risk Analysis and Management**: used to identify, analyze, develop plans to modify the risks, and to manage the risks on an ongoing basis.
- **Root Cause Analysis**: used to determine the underlying cause of performance variance.
- **Survey or Questionnaire**: used to determine expected value of a solution and its perceived performance.

8.2.7 Stakeholders

- **Domain Subject Matter Expert**: can identify risks and provide insights into data for analyzing solution performance.
- **Project Manager**: within a project, responsible for overall risk management and may participate in risk analysis for new or changed solutions.
- **Sponsor**: can identify risks, provide insights into data and the potential value of a solution. They will make decisions about the significance of expected versus actual solution performance.

8.2.8 Outputs

- **Solution Performance Analysis**: results of the analysis of measurements collected and recommendations to solve performance gaps and leverage opportunities to improve value.

8.3 Assess Solution Limitations

8.3.1 Purpose

The purpose of Assess Solution Limitations is to determine the factors internal to the solution that restrict the full realization of value.

8.3.2 Description

Assessing solution limitations identifies the root causes for under-performing and ineffective solutions and solution components.

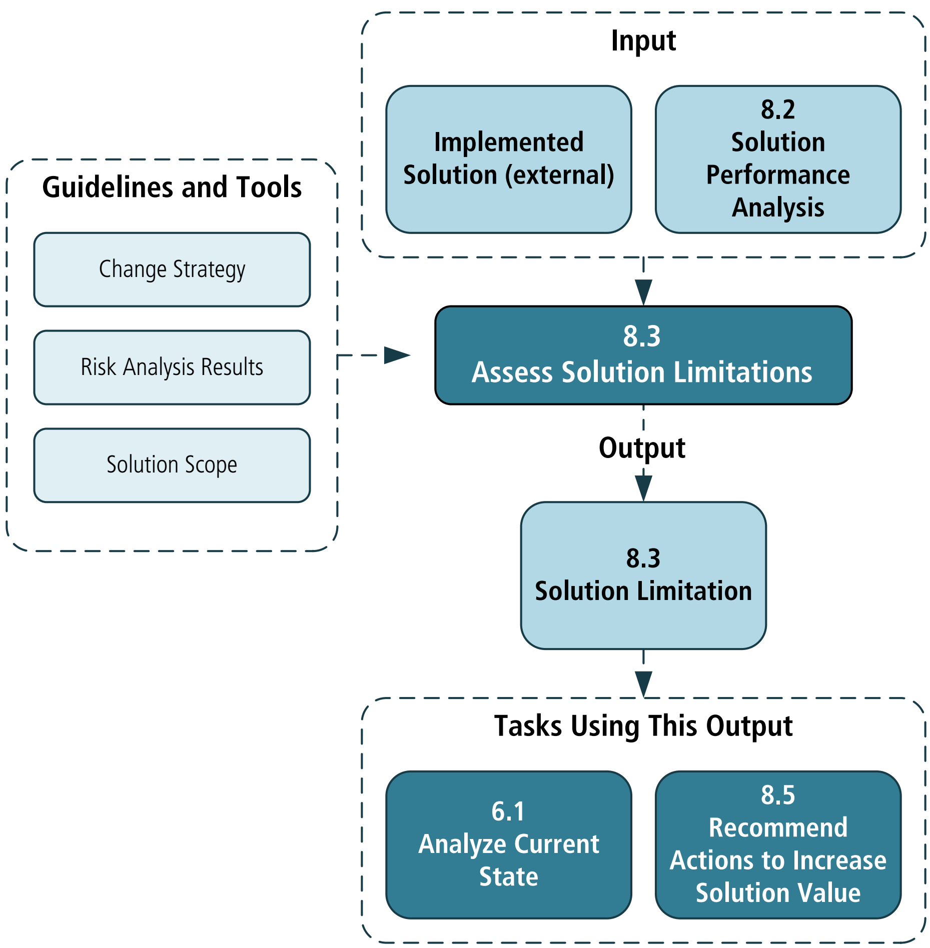
Assess Solution Limitations is closely linked to the task Assess Enterprise Limitations (p. 177). These tasks may be performed concurrently. If the solution has not met its potential value, business analysts determine which factors, both internal and external to the solution, are limiting value. This task focuses on the assessment of those factors internal to the solution.

This assessment may be performed at any point during the solution life cycle. It may occur on a solution component during its development, on a completed solution prior to full implementation, or on an existing solution that is currently working within an organization. Regardless of the timing, the assessment activities are similar and involve the same considerations.

8.3.3 Inputs

- **Implemented Solution (external)**: a solution that exists. The solution may or may not be in operational use; it may be a prototype. The solution must be in use in some form in order to be evaluated.
- **Solution Performance Analysis**: results of the analysis of measurements collected and recommendations to solve for performance gaps and leverage opportunities to improve value.

Figure 8.3.1: Assess Solution Limitations Input/Output Diagram



8.3.4 Elements

.1 Identify Internal Solution Component Dependencies

Solutions often have internal dependencies that limit the performance of the entire solution to the performance of the least effective component. Assessment of the overall performance of the solution or its components is performed in the tasks Measure Solution Performance (p. 166) and Analyze Performance Measures (p. 170). Business analysts identify solution components which have dependencies on other solution components, and then determine if there is anything about those dependencies or other components that limit solution performance and value realization.

.2 Investigate Solution Problems

When it is determined that the solution is consistently or repeatedly producing ineffective outputs, problem analysis is performed in order to identify the source of the problem.

Business analysts identify problems in a solution or solution component by examining instances where the outputs from the solution are below an acceptable level of quality or where the potential value is not being realized. Problems may be indicated by an inability to meet a stated goal, objective, or requirement, or may be a failure to realize a benefit that was projected during the tasks Define Change Strategy (p. 124) or Recommend Actions to Increase Solution Value (p. 182).

.3 Impact Assessment

Business analysts review identified problems in order to assess the effect they may have on the operation of the organization or the ability of the solution to deliver its potential value. This requires determining the severity of the problem, the probability of the re-occurrence of the problem, the impact on the business operations, and the capacity of the business to absorb the impact. Business analysts identify which problems must be resolved, which can be mitigated through other activities or approaches, and which can be accepted.

Other activities or approaches may include additional quality control measures, new or adjusted business processes, or additional support for exceptions to the desired outcome.

In addition to identified problems, business analysts assess risks to the solution and potential limitations of the solution. This risk assessment is specific to the solution and its limitations.

8.3.5 Guidelines and Tools

- **Change Strategy**: the change strategy used or in use to implement the potential value.

- **Risks Analysis Results**: the overall level of risk and the planned approach to modifying the individual risks.
- **Solution Scope**: the solution boundaries to measure and evaluate.

8.3.6 Techniques

- **Acceptance and Evaluation Criteria**: used both to indicate the level at which acceptance criteria are met or anticipated to be met by the solution and to identify any criteria that are not met by the solution.
- **Benchmarking and Market Analysis**: used to assess if other organizations are experiencing the same solution challenges and, if possible, determine how they are addressing it.
- **Business Rules Analysis**: used to illustrate the current business rules and the changes required to achieve the potential value of the change.
- **Data Mining**: used to identify factors constraining performance of the solution.
- **Decision Analysis**: used to illustrate the current business decisions and the changes required to achieve the potential value of the change.
- **Interviews**: used to help perform problem analysis.
- **Item Tracking**: used to record and manage stakeholder issues related to why the solution is not meeting the potential value.
- **Lessons Learned**: used to determine what can be learned from the inception, definition, and construction of the solution to have potentially impacted its ability to deliver value.
- **Risk Analysis and Management**: used to identify, analyze, and manage risks, as they relate to the solution and its potential limitations, that may impede the realization of potential value.
- **Root Cause Analysis**: used to identify and understand the combination of factors and their underlying causes that led to the solution being unable to deliver its potential value.
- **Survey or Questionnaire**: used to help perform problem analysis.

8.3.7 Stakeholders

- **Customer**: is ultimately affected by a solution, and therefore has an important perspective on its value. A customer may be consulted to provide reviews and feedback.
- **Domain Subject Matter Expert**: provides input into how the solution should perform and identifies potential limitations to value realization.

- **End User**: uses the solution, or is a component of the solution, and therefore contributes to the actual value realized by the solution in terms of solution performance. An end user may be consulted to provide reviews and feedback on areas such as workload and job satisfaction.
- **Regulator**: a person whose organization needs to be consulted about the planned and potential value of a solution, as that organization may constrain the solution, the degree to which actual value is realized, or when actual value is realized.
- **Sponsor**: responsible for approving the potential value of the solution, for providing resources to develop, implement and support the solution, and for directing enterprise resources to use the solution. The sponsor is also responsible for approving a change to potential value.
- **Tester**: responsible for identifying solution problems during construction and implementation; not often used in assessing an existing solution outside of a change.

8.3.8 Outputs

- **Solution Limitation**: a description of the current limitations of the solution including constraints and defects.

8.4 Assess Enterprise Limitations

8.4.1 Purpose

The purpose of Assess Enterprise Limitations is to determine how factors external to the solution are restricting value realization.

8.4.2 Description

Solutions may operate across various organizations within an enterprise, and therefore have many interactions and interdependencies. Solutions may also depend on environmental factors that are external to the enterprise. Enterprise limitations may include factors such as culture, operations, technical components, stakeholder interests, or reporting structures.

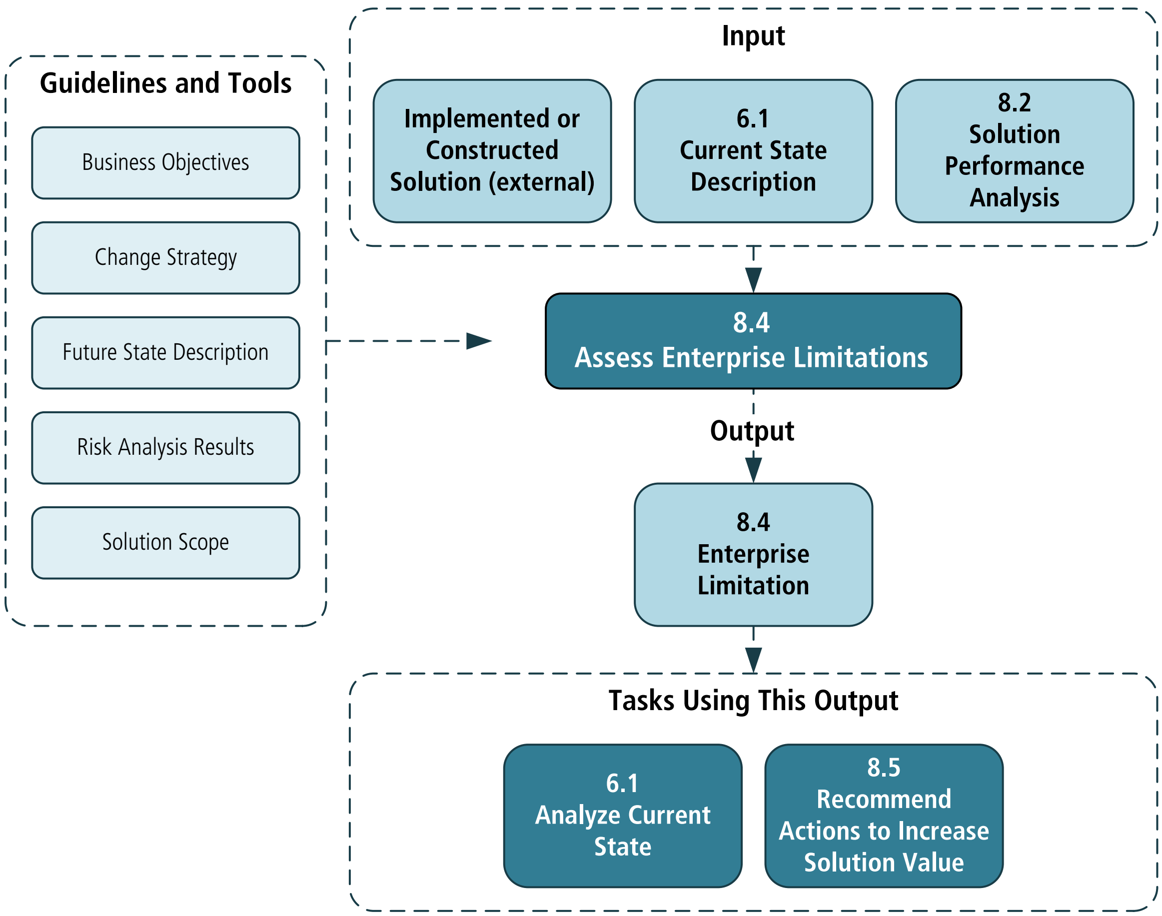
Assessing enterprise limitations identifies root causes and describes how enterprise factors limit value realization.

This assessment may be performed at any point during the solution life cycle. It may occur on a solution component during its development or on a completed solution prior to full implementation. It may also occur on an existing solution that is currently working within an organization. Regardless of the timing, the assessment activities are similar and require the same skills.

8.4.3 Inputs

- **Current State Description**: the current internal environment of the solution including the environmental, cultural, and internal factors influencing the solution limitations.
- **Implemented (or Constructed) Solution (external)**: a solution that exists. The solution may or may not be in operational use; it may be a prototype. The solution must be in use in some form in order to be evaluated.
- **Solution Performance Analysis**: results of the analysis of measurements collected and recommendations to solve performance gaps and leverage opportunities to improve value.

Figure 8.4.1: Assess Enterprise Limitations Input/Output Diagram



8.4.4 Elements

.1 Enterprise Culture Assessment

Enterprise culture is defined as the deeply rooted beliefs, values, and norms shared by the members of an enterprise. While these beliefs and values may not be directly visible, they drive the actions taken by an enterprise.

Business analysts perform cultural assessments to:

- identify whether or not stakeholders understand the reasons why a solution exists,
- ascertain whether or not the stakeholders view the solution as something beneficial and are supportive of the change, and
- determine if and what cultural changes are required to better realize value from a solution.

The enterprise culture assessment evaluates the extent to which the culture can accept a solution. If cultural adjustments are needed to support the solution, the assessment is used to judge the enterprise's ability and willingness to adapt to these cultural changes.

Business analysts also evaluate internal and external stakeholders to:

- gauge understanding and acceptance of the solution,
- assess perception of value and benefit from the solution, and
- determine what communication activities are needed to ensure awareness and understanding of the solution.

.2 Stakeholder Impact Analysis

A stakeholder impact analysis provides insight into how the solution affects a particular stakeholder group.

When conducting stakeholder impact analysis, business analysts consider:

- **Functions**: the processes in which the stakeholder uses the solution, which include inputs a stakeholder provides into the process, how the stakeholder uses the solution to execute the process, and what outputs the stakeholder receives from the process.
- **Locations**: the geographic locations of the stakeholders interacting with the solution. If the stakeholders are in disparate locations, it may impact their use of the solution and the ability to realize the value of the solution.
- **Concerns**: the issues, risks, and overall concerns the stakeholders have with the solution. This may include the use of the solution, the perceptions of the value of the solution, and the impact the solution has on a stakeholder's ability to perform necessary functions.

.3 Organizational Structure Changes

There are occasions when business analysts assess how the organization's structure is impacted by a solution.

The use of a solution and the ability to adopt a change can be enabled or blocked by formal and informal relationships among stakeholders. The reporting structure may be too complex or too simple to allow a solution to perform effectively. Assessing if the organizational hierarchy supports the solution is a key activity. On occasion, informal relationships within an organization, whether alliances, friendships, or matrix-reporting, impact the ability of a solution to deliver potential value. Business analysts consider these informal relationships in addition to the formal structure.

.4 Operational Assessment

The operational assessment is performed to determine if an enterprise is able to adapt to or effectively use a solution. This identifies which processes and tools within the enterprise are adequately equipped to benefit from the solution, and if sufficient and appropriate assets are in place to support it.

When conducting an operational assessment, business analysts consider:

- policies and procedures,
- capabilities and processes that enable other capabilities,
- skill and training needs,
- human resources practices,
- risk tolerance and management approaches, and
- tools and technology that support a solution.

8.4.5 Guidelines and Tools

- **Business Objectives**: are considered when measuring and determining solution performance.
- **Change Strategy**: the change strategy used or in use to implement the potential value.
- **Future State Descriptions**: boundaries of the proposed new, removed, or modified components of the enterprise, as well as the potential value expected from the future state.
- **Risk Analysis Results**: the overall level of risk and the planned approach to modifying the individual risks.
- **Solution Scope**: the solution boundaries to measure and evaluate.

8.4.6 Techniques

- **Benchmarking and Market Analysis**: used to identify existing solutions and enterprise interactions.
- **Brainstorming**: used to identify organizational gaps or stakeholder concerns.
- **Data Mining**: used to identify factors constraining performance of the solution.
- **Decision Analysis**: used to assist in making an optimal decision under conditions of uncertainty and may be used in the assessment to make decisions about functional, technical, or procedural gaps.
- **Document Analysis**: used to gain an understanding of the culture, operations, and structure of the organization.

- **Interviews**: used to identify organizational gaps or stakeholder concerns.
- **Item Tracking**: used to ensure that issues are not neglected or lost and that issues identified by assessment are resolved.
- **Lessons Learned**: used to analyze previous initiatives and the enterprise interactions with the solutions.
- **Observation**: used to witness the enterprise and solution interactions to identify impacts.
- **Organizational Modelling**: used to ensure the identification of any required changes to the organizational structure that may have to be addressed.
- **Process Analysis**: used to identify possible opportunities to improve performance.
- **Process Modelling**: used to illustrate the current business processes and/or changes that must be made in order to achieve the potential value of the solution.
- **Risk Analysis and Management**: used to consider risk in the areas of technology (if the selected technological resources provide required functionality), finance (if costs could exceed levels that make the change salvageable), and business (if the organization will be able to make the changes necessary to attain potential value from the solution).
- **Roles and Permissions Matrix**: used to determine roles and associated permissions for stakeholders, as well as stability of end users.
- **Root Cause Analysis**: used to determine if the underlying cause may be related to enterprise limitations.
- **Survey or Questionnaire**: used to identify organizational gaps or stakeholder concerns.
- **SWOT Analysis**: used to demonstrate how a change will help the organization maximize strengths and minimize weaknesses, and to assess strategies developed to respond to identified issues.
- **Workshops**: used to identify organizational gaps or stakeholder concerns.

8.4.7 Stakeholders

- **Customer**: people directly purchasing or consuming the solution who may interact with the organization in the use of the solution.
- **Domain Subject Matter Expert**: provides input into how the organization interacts with the solution and identifies potential limitations.
- **End User**: people who use a solution or who are a component of the solution. Users could be customers or people who work within the organization.
- **Regulator**: one or many governmental or professional entities that ensure adherence to laws, regulations, or rules; may have unique input to the organizational assessment, as relevant regulations must be included in the

requirements. There may be laws and regulations that must be complied with prior to (or as a result of) a planned or implemented change.

- **Sponsor**: authorizes and ensures funding for a solution delivery, and champions action to resolve problems identified in the organizational assessment.

8.4.8 Outputs

- **Enterprise Limitation**: a description of the current limitations of the enterprise including how the solution performance is impacting the enterprise.

8.5 Recommend Actions to Increase Solution Value

8.5.1 Purpose

The purpose of Recommend Actions to Increase Solution Value is to understand the factors that create differences between potential value and actual value, and to recommend a course of action to align them.

8.5.2 Description

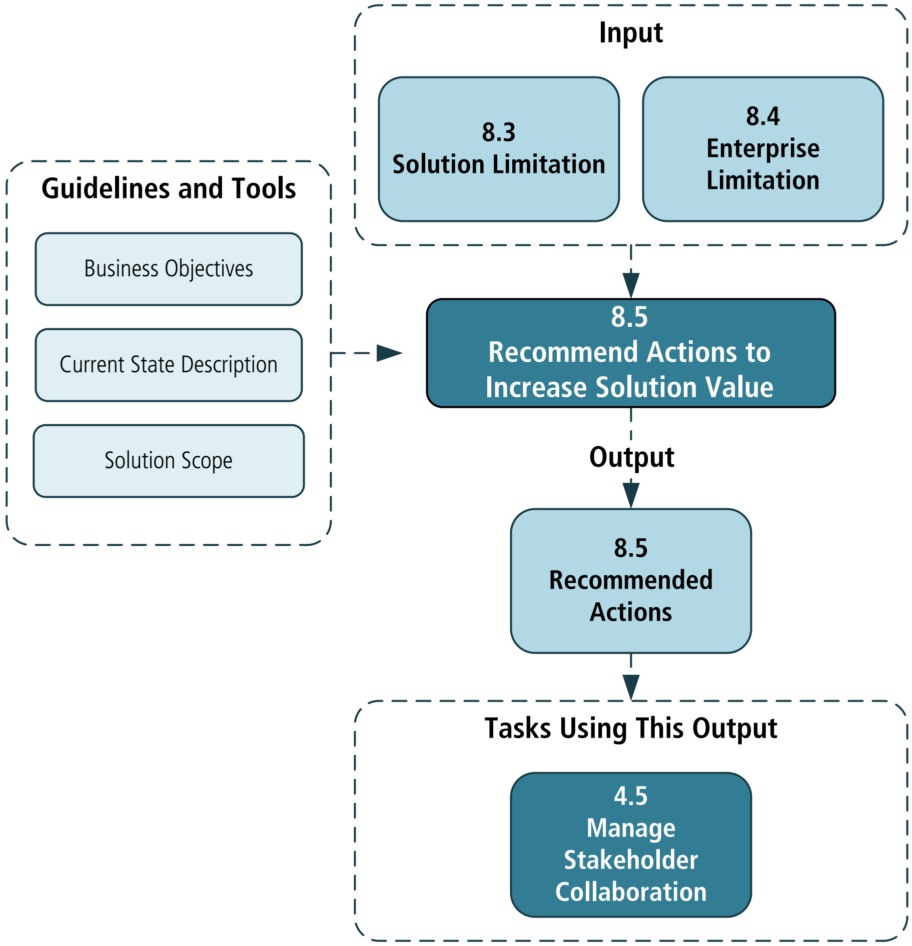
The various tasks in the Solution Evaluation knowledge area help to measure, analyze, and determine causes of unacceptable solution performance. The task Recommend Actions to Increase Solution Value (p. 182), focuses on understanding the aggregate of the performed assessments and identifying alternatives and actions to improve solution performance and increase value realization.

Recommendations generally identify how a solution should be replaced, retired, or enhanced. They may also consider long-term effects and contributions of the solution to stakeholders. They may include recommendations to adjust the organization to allow for maximum solution performance and value realization.

8.5.3 Inputs

- **Enterprise Limitation**: a description of the current limitations of the enterprise including how the solution performance is impacting the enterprise.
- **Solution Limitation**: a description of the current limitations of the solution including constraints and defects.

Figure 8.5.1: Recommend Actions to Increase Solution Value Input/Output Diagram



8.5.4 Elements

.1 Adjust Solution Performance Measures

In some cases, the performance of the solution is considered acceptable but may not support the fulfillment of business goals and objectives. An analysis effort to identify and define more appropriate measures may be required.

.2 Recommendations

While recommendations often describe ways to increase solution performance, this is not always the case. Depending on the reason for lower than expected performance, it may be reasonable to take no action, adjust factors that are external to the solution, or reset expectations for the solution.

Some common examples of recommendations that a business analyst may make include:

- **Do Nothing**: is usually recommended when the value of a change is low relative to the effort required to make the change, or when the risks of change significantly outweigh the risks of remaining in the current state. It may also be impossible to make a change with the resources available or in the allotted time frame.

- **Organizational Change**: is a process for managing attitudes about, perceptions of, and participation in the change related to the solution. Organizational change management generally refers to a process and set of tools for managing change at an organizational level. The business analyst may help to develop recommendations for changes to the organizational structure or personnel, as job functions may change significantly as the result of work being automated. New information may be made available to stakeholders and new skills may be required to operate the solution. Possible recommendations that relate to organizational change include:
  - automating or simplifying the work people perform. Relatively simple tasks are prime candidates for automation. Additionally, work activities and business rules can be reviewed and analyzed to determine opportunities for re-engineering, changes in responsibilities, and outsourcing.
  - improving access to information. Change may provide greater amounts of information and better quality of information to staff and decision makers.
- **Reduce Complexity of Interfaces**: interfaces are needed whenever work is transferred between systems or between people. Reducing their complexity can improve understanding.
- **Eliminate Redundancy**: different stakeholder groups may have common needs that can be met with a single solution, reducing the cost of implementation.
- **Avoid Waste**: the aim of avoiding waste is to completely remove those activities that do not add value and minimize those activities that do not contribute to the final product directly.
- **Identify Additional Capabilities**: solution options may offer capabilities to the organization above and beyond those identified in the requirements. In many cases, these capabilities are not of immediate value to the organization but have the potential to provide future value, as the solution may support the rapid development or implementation of those capabilities if they are required (for example, a software application may have features that the organization anticipates using in the future).
- **Retire the Solution**: it may be necessary to consider the replacement of a solution or solution component. This may occur because technology has reached the end of its life, services are being insourced or outsourced, or the solution is not fulfilling the goals for which it was created.
- Some additional factors that may impact the decision regarding the replacement or retirement of a solution include:
  - **ongoing cost versus initial investment**: it is common for the existing solution to have increasing costs over time, while alternatives have a higher investment cost upfront but lower maintenance costs.

- **opportunity cost**: represents the potential value that could be realized by pursuing alternative courses of action.
- **necessity**: most solution components have a limited lifespan (due to obsolescence, changing market conditions, and other causes). After a certain point in the life cycle it will become impractical or impossible to maintain the existing component.
- **sunk cost**: describes the money and effort already committed to an initiative. The psychological impact of sunk costs may make it difficult for stakeholders to objectively assess the rationale for replacement or elimination, as they may feel reluctant to "waste" the effort or money already invested. As this investment cannot be recovered, it is effectively irrelevant when considering future action. Decisions should be based on the future investment required and the future benefits that can be gained.

8.5.5 Guidelines and Tools

- **Business Objectives**: are considered in evaluating, measuring, and determining solution performance.
- **Current State Description**: provides the context within which the work needs to be completed. It can be used to assess alternatives and better understand the potential increased value that could be delivered. It can also help highlight unintended consequences of alternatives that may otherwise remain undetected.
- **Solution Scope**: the solution boundaries to measure and evaluate.

8.5.6 Techniques

- **Data Mining**: used to generate predictive estimates of solution performance.
- **Decision Analysis**: used to determine the impact of acting on any of the potential value or performance issues.
- **Financial Analysis**: used to assess the potential costs and benefits of a change.
- **Focus Groups**: used to determine if solution performance measures need to be adjusted and used to identify potential opportunities to improve performance.
- **Organizational Modelling**: used to demonstrate potential change within the organization's structure.
- **Prioritization**: used to identify relative value of different actions to improve solution performance.
- **Process Analysis**: used to identify opportunities within related processes.
- **Risk Analysis and Management**: used to evaluate different outcomes under specific conditions.
- **Survey or Questionnaire**: used to gather feedback from a wide variety of stakeholders to determine if value has been met or exceeded, if the metrics are

still valid or relevant in the current context, and what actions might be taken to improve the solution.

8.5.7 Stakeholders

- **Customer**: people directly purchasing or consuming the solution and who may interact with the organization in the use of the solution.
- **Domain Subject Matter Expert**: provides input into how to change the solution and/or the organization in order to increase value.
- **End User**: people who use a solution or who are a component of the solution. Users could be customers or people who work within the organization.
- **Regulator**: one or many governmental or professional entities that ensure adherence to laws, regulations, or rules. Relevant regulations must be included in requirements.
- **Sponsor**: authorizes and ensures funding for implementation of any recommended actions.

8.5.8 Outputs

- **Recommended Actions**: recommendation of what should be done to improve the value of the solution within the enterprise.